

# DSA Basics — Beginner Notes PDF

Based on today's class topics | Arrays + Basic DSA Concepts

This PDF is prepared according to the topics you shared from today's DSA basics session. It is kept at a beginner level and explains each concept in a simple way, with examples and basic Python list functions wherever arrays were discussed.

## ## Topics Covered in Today's Session

- What is Data Structure?
- What is Algorithm?
- Examples of Data Structures
- Examples of Algorithms
- What is an Array?
- Functions / operations of Array
- What is pop() and how many types of pop usage are there?
- Which function is used to insert an element at the end?
- Which function is used to remove a specific element?
- Which function is used to remove the last element?

## ## 1) What is Data Structure?

A Data Structure is a way of storing and organizing data in a computer so that it can be used efficiently. In simple words, it tells us how data is arranged and how we can access or modify it easily.

For example, if you have marks of students, you can store them in an array or list. If you have browser history, it behaves like a stack. So different problems use different data structures.

## ## 2) What is Algorithm?

An Algorithm is a step-by-step procedure used to solve a problem. It is like a recipe: it gives the exact steps needed to get the result.

- Take input
- Process the data step by step
- Produce the final answer

## ## 3) Difference between Data Structure and Algorithm

| Data Structure                   | Algorithm                                             |
|----------------------------------|-------------------------------------------------------|
| How data is stored and organized | How a problem is solved step by step                  |
| Examples: Array, Stack, Queue    | Examples: Searching, Sorting, Largest element finding |

## ## 4) Examples of Data Structures

- Array / List

- Stack
- Queue
- Linked List
- Tree
- Graph

## ## 5) Examples of Algorithms

- Searching algorithm
- Sorting algorithm
- Largest element algorithm
- Sum of array algorithm
- Palindrome checking algorithm

## ## 6) What is an Array?

An Array is a data structure used to store multiple values in a single variable. In beginner Python learning, arrays are usually explained using lists.

```
arr = [10, 20, 30, 40]
print(arr[0])    # 10
print(arr[1])    # 20
```

## ## 7) Functions / Operations of Array

| Operation | Meaning                       | Example            |
|-----------|-------------------------------|--------------------|
| Insert    | Add a new element             | append(), insert() |
| Delete    | Remove an element             | pop(), remove()    |
| Access    | Get an element using index    | arr[0]             |
| Update    | Change a value                | arr[1] = 99        |
| Traverse  | Visit all elements one by one | for x in arr:      |

## ## 8) Which function is used to insert an element at the end?

We use append() to add an element at the end of a list.

```
arr = [10, 20, 30]
arr.append(40)
print(arr)    # [10, 20, 30, 40]
```

## ## 9) What is insert()?

insert(index, value) is used to add an element at a specific position.

```
arr = [10, 20, 30]
arr.insert(1, 99)
print(arr)    # [10, 99, 20, 30]
```

## ## 10) What is pop()?

pop() removes an element from a list.

- `pop()` removes the last element
- `pop(index)` removes the element at a specific index

```
arr = [10, 20, 30, 40]
arr.pop()
print(arr)    # [10, 20, 30]

arr = [10, 20, 30, 40]
arr.pop(1)
print(arr)    # [10, 30, 40]
```

## ## 11) Which function is used to remove a specific element?

We use `remove(value)` to remove a specific value from a list.

```
arr = [10, 20, 30, 40]
arr.remove(30)
print(arr)    # [10, 20, 40]
```

## ## 12) Which function is used to remove the last element?

We commonly use `pop()` to remove the last element.

```
arr = [5, 10, 15, 20]
arr.pop()
print(arr)    # [5, 10, 15]
```

## ## 13) Difference between `append()`, `insert()`, `remove()`, and `pop()`

| Function                  | Purpose                    |
|---------------------------|----------------------------|
| <code>append(x)</code>    | Adds x at the end          |
| <code>insert(i, x)</code> | Adds x at index i          |
| <code>remove(x)</code>    | Removes value x            |
| <code>pop()</code>        | Removes last element       |
| <code>pop(i)</code>       | Removes element at index i |

## ## 14) Basic Combined Example

```
arr = [10, 20, 30]

arr.append(40)
print("After append:", arr)

arr.insert(1, 99)
print("After insert:", arr)

arr.remove(20)
print("After remove:", arr)

arr.pop(2)
print("After pop(index):", arr)

arr.pop()
print("After pop():", arr)
```

## ## 15) Viva Questions from Today's DSA Basics

- What is a Data Structure?

- What is an Algorithm?
- Give examples of Data Structures.
- Give examples of Algorithms.
- What is an Array?
- What does append() do?
- What is the use of remove()?
- What is the use of pop()?
- What is the difference between pop() and remove()?

## ## 16) Quick Revision Box

| Topic          | Key point                                        |
|----------------|--------------------------------------------------|
| Data Structure | Way of storing and organizing data               |
| Algorithm      | Step-by-step procedure to solve a problem        |
| Array          | Stores multiple values in one variable           |
| append()       | Adds element at the end                          |
| insert()       | Adds element at a specific position              |
| remove()       | Removes a specific value                         |
| pop()          | Removes last element or element at a given index |